

X4Pro. News-2026.

1. Code reac1.py: universal retrieval and plotting of EXFOR data having one-string Reaction-codes.

Examples of newly supported quantities:

- CSP Partial cross section data
- CST Temperature dependent cross section
- ETA Average neutron yield per nonelastic event
- NU Fission-neutron yield, nu-bar
- TKE Total kinetic energy

2. Plotting partial cross sections from local EXFOR and remote ENDF databases.

- CSP vs ENDF:MT51 - Production of a neutron, with residual in the 1st excited state

Code reac1.py help:

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| Program: reac1.py, ver.2026-08-14 |
| Author:  V.Zerkin, Vienna, 2021-2026 |
| Purpose: Retrieve and plot any type of |
|          data from local EXFOR database |
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-----Help-----
Purpose: Retrieve and plot any type EXFOR data from local EXFOR database

Introduction:
* computational data are presented as function y=f(x1,x2,..)
  where f is measured quantity, xi are independent variables:
  EN  - incident energy
  ANG - angle of outgoing particle
  E2  - energy of outgoing particle
  EVL - energy level
  etc.

Algorithm:
* user defines EXFOR Reaction-code for data search, x-variable by number
* selects x-variable by number (default: x1)
* other variables will be used as parameter for grouping of data points

Usage:
$ python [{flag}] x4update.py [{option|reaction}]
* flag: see all Python flags: $ python --help
  -B      don't write .pyc files on import
* option:
  -help      print this help-text and exit (also --help)
  -o:<file>  output file name
  -x:x?     set x variable: "x1" to "x5", e.g. -x:x2, default: -x:x1
  -x?fam:<str> filter for "families" of x? variable, e.g. -x2fam:"LVL;EXC"
  -x?min:<num> filter out values of x? variable, e.g. -x1min:6e6
  -x?max:<num> filter out values of x? variable, e.g. -x1max:30e6
  -nogrp     avoid grouping datasets by reaction-codes on the plot
  -fx:<num>  set factor for x-values, e.g. -fx:1e6 sets "MeV" on X-Axis
  -fy:<num>  set factor for y-values, e.g. -fy:1e-3 sets "mb/sr" on Y-Axis
plotting:
  -xlog      logarothmic scale of X-Axis
  -ylog      logarothmic scale of Y-Axis
  -lines     connect points on the plot by lines
  -sym       draw border of a symbol of a data point on the plot
  -annot:x,y,txt annotation on top of the plot;
              x and y should be given in units of the plot;
              text can include html tags: <b>, <sup>
* reaction:  file name should not start with sign "-"
  <reaction> full EXFOR Reaction-code, e.g. "8-0-16(N,EL)8-0-16,,DA"

Examples.
1) SIG(Ei) Cross sections
$ python -B reac1.py "13-AL-27(N,A)11-NA-24,,SIG"

2) DA(Ei) Angular distributions
$ python3 -B reac1.py -o:da1e -x:x1 \
  -x1max:6.5e6 -x2min:160 -x2max:170 \
  -lines -sym -ylog -fx:1e3 -fy:1e-3 \
  -annot:"2500,250,<b>DA(E)</b> <sup>14</sup>N(a,a')" \
  "7-N-14(A,EL)7-N-14,,DA">da1e.tto

3) DE(Eo) Emission spectra
$ python -B reac1.py -o:de1 -x:x2 \
  -x1min:6e6 -fx:1e6 -fy:1e-9 -lines -ylog \
  "90-TH-232(N,X)0-NN-1,,DE"

4) DAE(Eo) Double differential cross section
$ python -B reac1.py -o:de1 -x:x2 \
  -x1min:6e6 -fx:1e6 -fy:1e-9 -lines -ylog \
  "90-TH-232(N,X)0-NN-1,,DE"

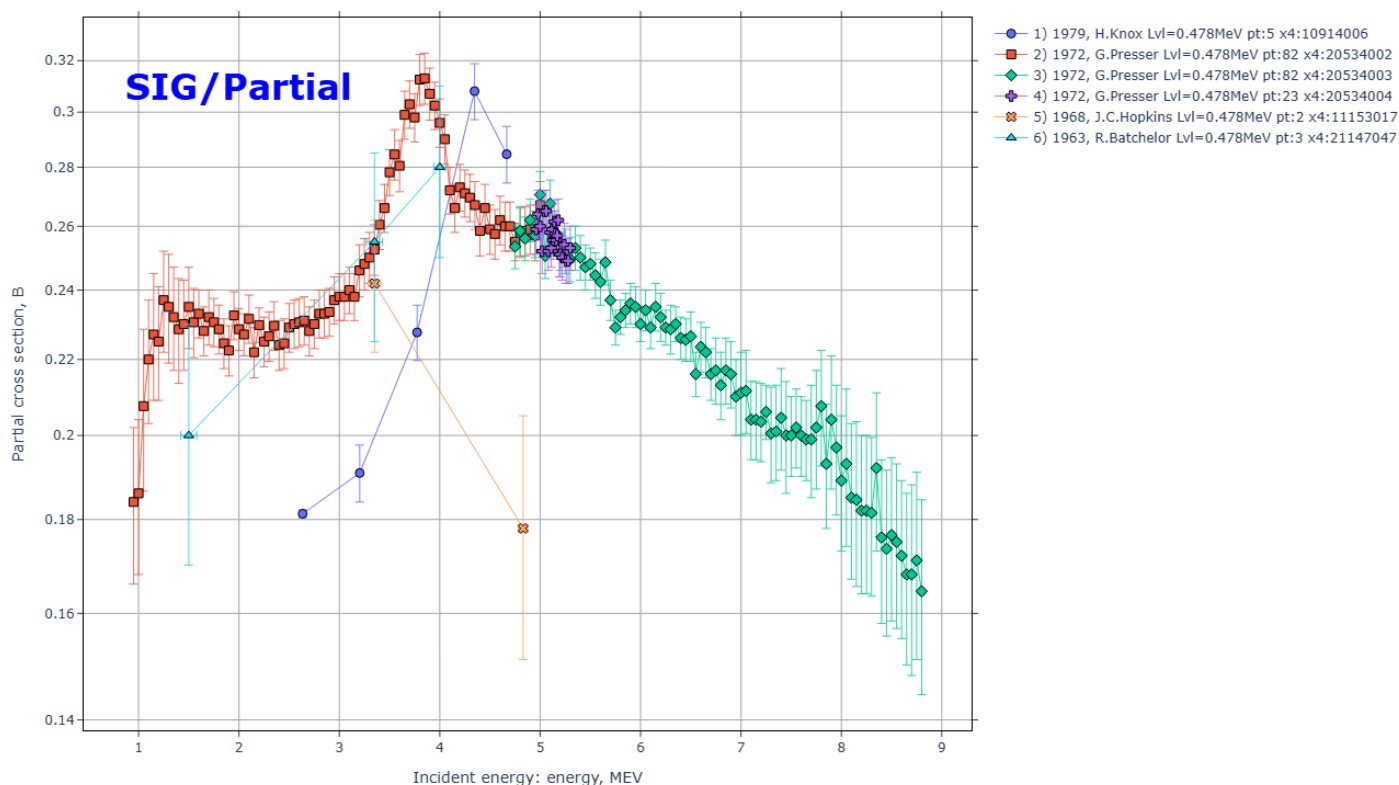
5) TKE(Mass) Total Kinetic Energy
$ python -B reac1.py -o:tke1 -x:x2 \
  -fy:1e6 -lines -sym \
  -annot:"80,180,<b>TKE</b>" \
  "92-U-235(N,F)MASS,PRES,KE,LF+HF" \
  "92-U-235(N,F)MASS,PRES,KE,LF+HF,MXW"

6) ETA(Ri) Neutron yield (Eta)
$ python -B reac1.py -o:eta1 -x:x1 -x1max:4 \
  -lines -sym -ylog \
  -annot:"0.75,3.25,<b>ETA</b>" \
  "92-U-235(N,ABS),,ETA">eta1.tto
```

part1-8-reac/ EXFOR universal: SIG, SIG-T, SIG-PAR, DA, DE, DA-PAR, FY, NUBAR, TKE, ETA

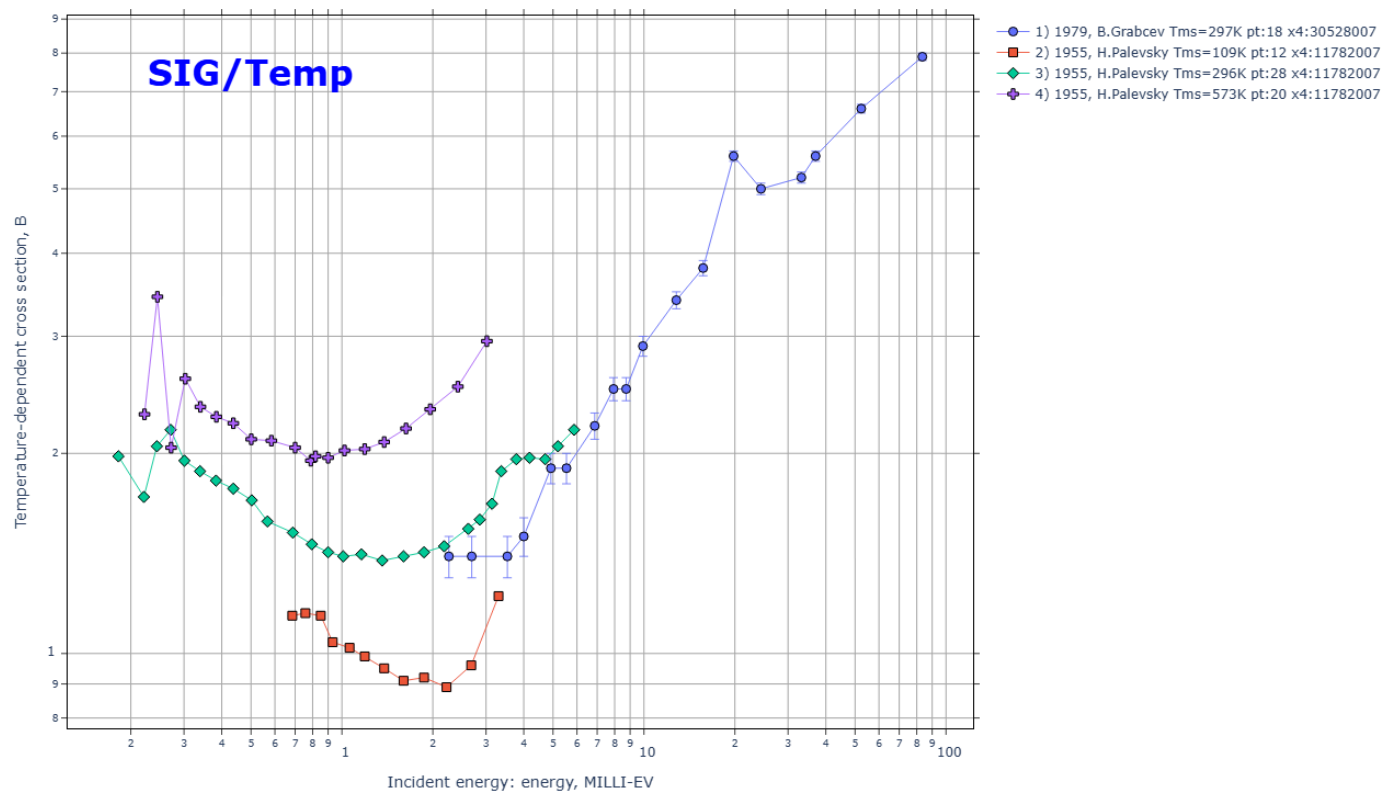
1) part1-8-reac/out00/sig1p.html.png CSP Partial cross section data

Reaction:3-LI-7(N,INL)3-LI-7,PAR,SIG Quantity:CSP:y=Data(En,Lvl) Datasets:6 Points:197
X4Pro, by V.Zerkin, NRDC, 2026, ver.2026-08-14 //running:2026-08-21 12:04:21



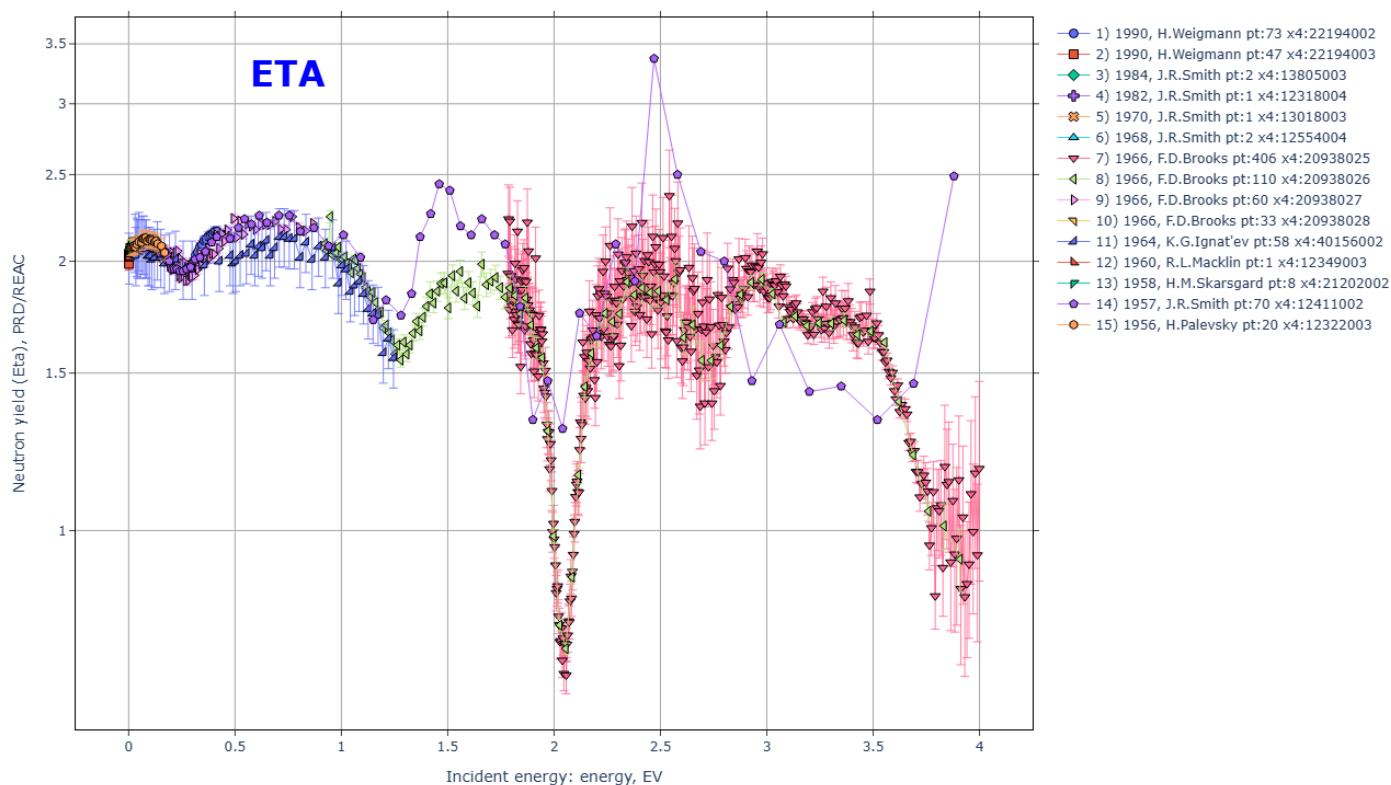
2) part1-8-reac/out00/cst1.html.png CST Temperature dependent cross section

Reaction:82-PB-0(N,TOT),,SIG/TMP Quantity:CST:y=Data(En,Tms) Datasets:4 Points:78
X4Pro, by V.Zerkin, NRDC, 2026, ver.2026-08-14 //running:2026-08-21 12:04:08



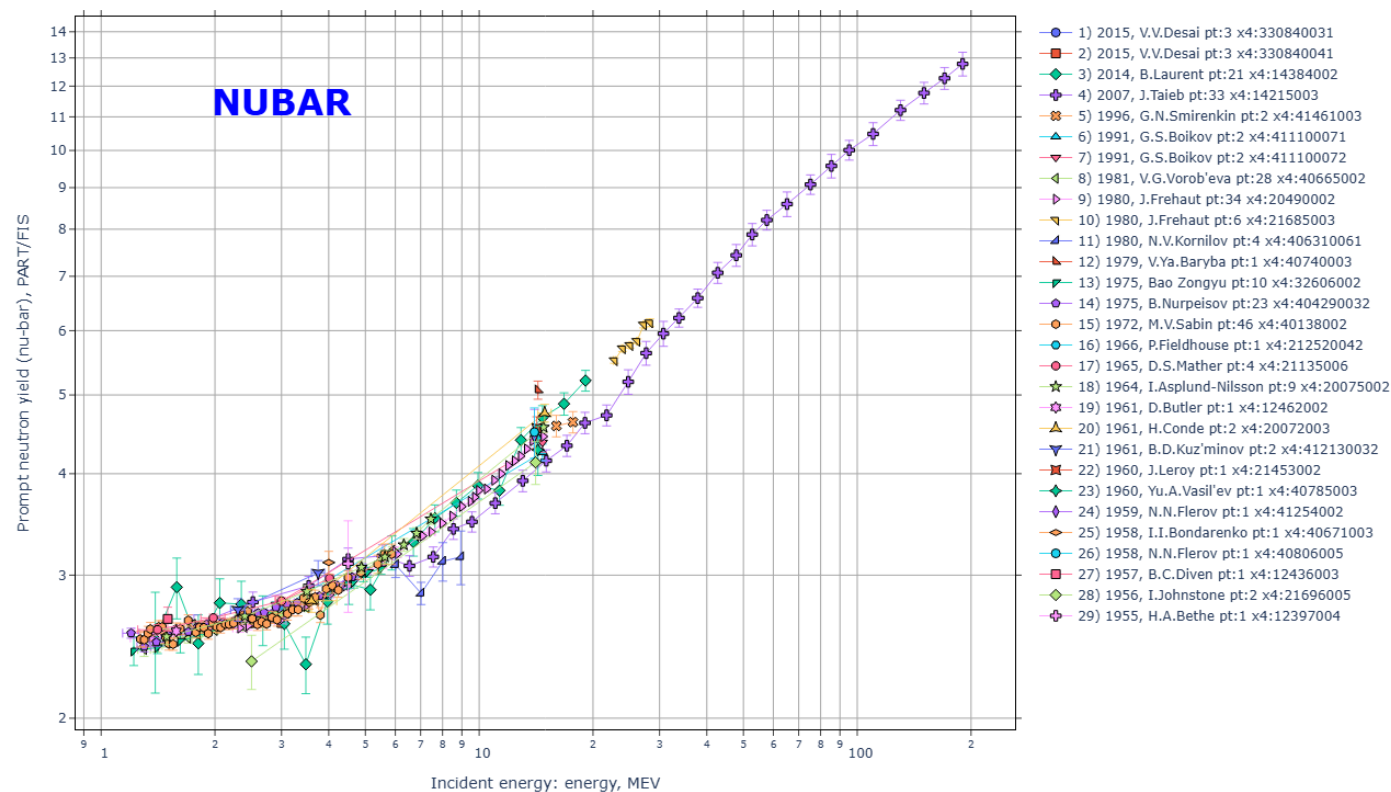
3) part1-8-reac/out00/eta1.html.png ETA Average neutron yield per nonelastic event

Reaction: 92-U-235(N,ABS),,ETA Quantity: NU:y=Data(En) Datasets: 15 Points: 892
X4Pro, by V.Zerkin, NRDC, 2026, ver.2026-08-14 //running:2026-08-21 12:04:18



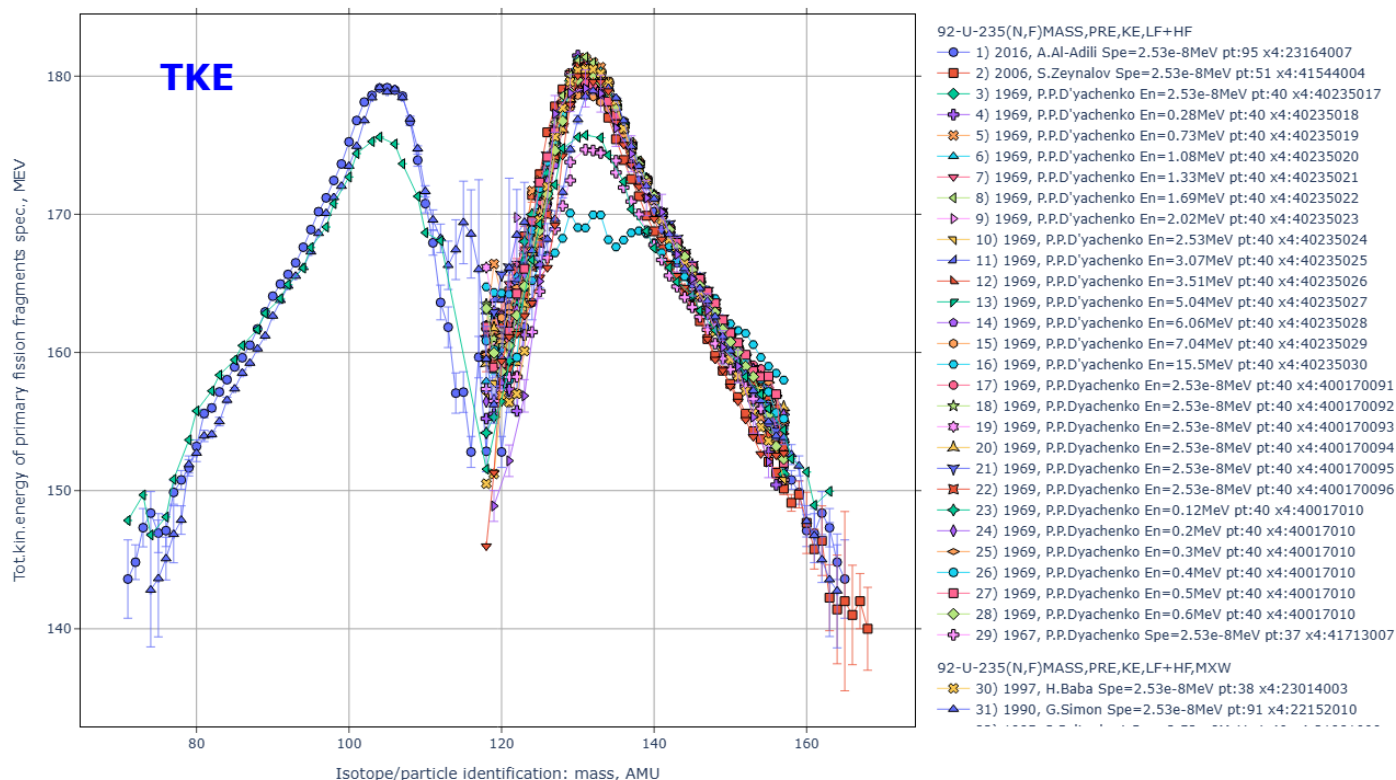
4) part1-8-reac/out00/nu1.html.png NU Fission-neutron yield, nu-bar

Reaction: 92-U-238(N,F),PR,NU Quantity: NU:y=Data(En) Datasets: 29 Points: 246
X4Pro, by V.Zerkin, NRDC, 2026, ver.2026-08-14 //running:2026-08-21 12:04:11



5) part1-8-reac/out00/tke1.html.png TKE Total kinetic energy

Reaction: 92-U-235(N,F)MASS,PKE,LF+HF Quantity: E:y=Data(Spe,Zam) Datasets: 34 Points: 1467
X4Pro, by V.Zerkin, NRDC, 2026, ver.2026-08-14 //running:2026-08-21 12:04:15



part2-8-sig1par/ EXFOR/ENDF partial cross sections CSP:MT51

6) part2-8-sig1par/out00/Li7nin1.html.png Production of a neutron, with residual in the 1st excited state

EXFOR/ENDF cross sections SIG(E): Li-7(n,n')par,sig EXFOR-datasets: 7
X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2026, ver.2026-08-10 //run:2026-08-21 12:08:56

